

COMPUTER NETWORK HANDBOOK

— PART - 2 —

Page 1 - Network Devices

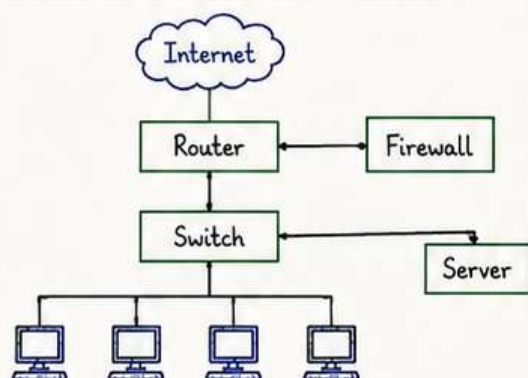
→ Network devices are hardware devices that connect different devices in a network and help in communication, routing, filtering and delivering data.

Why Network Devices ?

- Enable communication between devices.
- Manage and control data flow.
- Improve security and performance.
- Form the backbone of any network.

Device	Works on Layer	Main Purpose / Function	Example / Use
Hub	1	Connects multiple devices and broadcasts data to all ports.	Old wired networks (rarely used now)
Switch	2	Connects devices and forwards data to the correct port (MAC address).	Office LAN, Home Wi-Fi networks
Bridge	2	Connects two LAN segments and filters data.	Connecting two LANs
Repeater	1	Regenerates and amplifies signals to extend the network.	Long distance network (signal booster)
Router	3, 4, 7	Routes data between different networks using IP address.	Home router, ISP routing
Gateway	3, 4	Acts as an entry/exit point between different networks using different protocols.	Connecting dissimilar networks
Firewall		Monitors and controls incoming and outgoing network traffic based on rules.	Network security, Protection from attacks

Small Office Network Diagram



Key Points

- Hub broadcasts data to all devices.
- Switch sends data only to the correct device.
- Bridge reduces traffic between LAN segments.
- Repeater extends the reach of a network.
- Router connects different networks.
- Gateway translates between different protocols.
- Firewall protects the network from unauthorized access.

In Short : Network devices work together to build a reliable, fast and secure network.

COMPUTER NETWORK HANDBOOK

— PART - 2 —

Page 2 - Ports & Common Network Protocols

→ A **Port** is a logical endpoint used by applications to send and receive data over a network.
Each port is identified by a port number (0 - 65535).

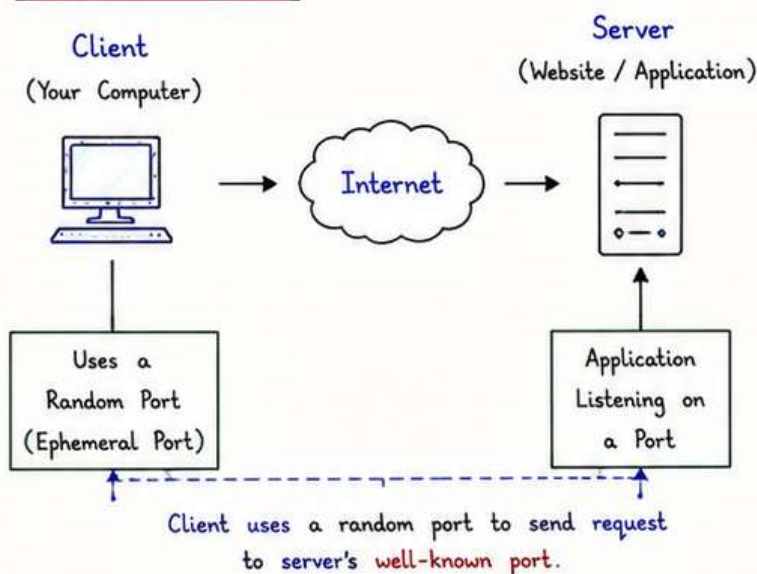
Types of Ports

- Well-known Ports : 0 - 1023
- Registered Ports : 1024 - 49151
- Dynamic / Private Ports : 49152 - 65535

Common Ports

Port Number	Protocol	Service / Protocol	Description	Used By
80	TCP	HTTP	Hyper Text Transfer Protocol	Web Browsers
443	TCP	HTTPS	HTTP Secure (SSL/TLS)	Secure Web Browsers
21	TCP	FTP	File Transfer Protocol	File Transfer
22	TCP	SSH	Secure Shell	Remote Login
25	TCP	SMTP	Simple Mail Transfer Protocol	Sending E-mails
53	UDP/TCP	DNS	Domain Name System	Domain Resolution
110	TCP	POP3	Post Office Protocol v3	Receiving E-mails
143	TCP	IMAP	Internet Message Access Protocol	Receiving E-mails
3306	TCP	MySQL	MySQL Database	Database Connection
5432	TCP	PostgreSQL	PostgreSQL Database	Database Connection

How Ports Work ?



Key Points

- A port helps the server know which **application** should receive the data.
- Multiple applications can run on the same server using **different ports**.
- 0 - 1023 ports require special permissions (**admin/root**).
- Most client applications use dynamic ports (49152 - 65535).

In Short : Ports act like **doors** in a server.

Requests come to the **correct door (port)** and are handled by the **right application**.

Page 3 - ARP, DHCP & NAT

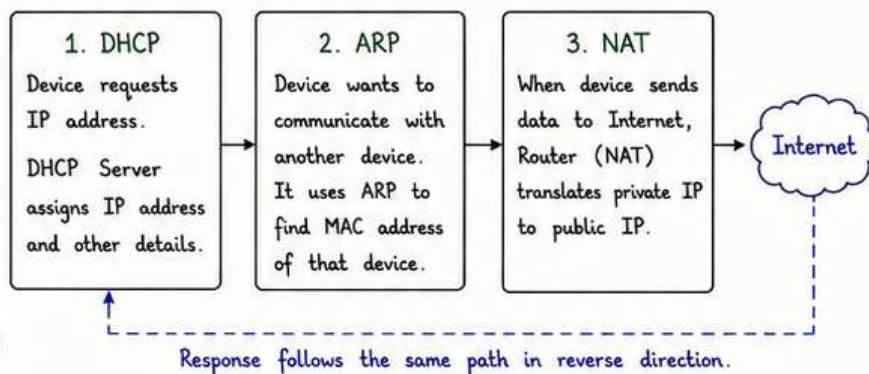
→ These are important protocols and mechanisms that help devices in a network to get an IP address, find MAC address and access the Internet.

What we will cover ?

- ARP → IP to MAC address mapping
- DHCP → Assigns IP address to devices
- NAT → Translates private IP to public IP
- Private vs Public IP

Topic	Full Form	Purpose	Works On
ARP	Address Resolution Protocol	Finds MAC address of a device knowing its IP address.	Local Network (LAN)
DHCP	Dynamic Host Configuration Protocol	Automatically assigns IP address and other network details to devices.	Network Server
NAT	Network Address Translation	Converts private IP address to public IP address before sending data to Internet.	Router / Firewall
Private IP	-	Used inside local network. Not routable on Internet.	RFC 1918
Public IP	-	Used on Internet. Routable and unique worldwide.	IANA

How it all works together ?

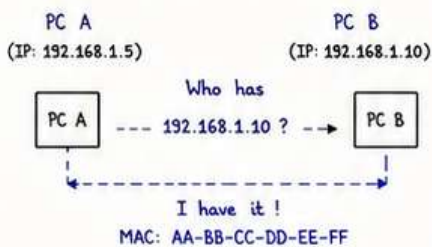


Private vs Public IP

Private IP	Public IP
Used inside LAN	Used on Internet
Not routable	Routable
Example: 10.0.0.0 to 10.255.255.255 172.16.0.0 to 172.31.255.255 192.168.0.0 to 192.168.255.255	Example: 8.8.8.8 1.1.1.1

1. ARP (Address Resolution Protocol)

Converts IP address to MAC address.



- ARP Request is broadcast in LAN.
- ARP Reply is unicast with MAC address.

2. DHCP (Dynamic Host Configuration Protocol)

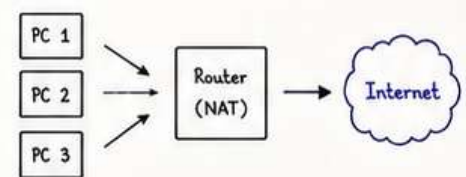
Provides IP address and other details.



- Assigns IP address
- Subnet Mask
- Default Gateway
- DNS Server

3. NAT (Network Address Translation)

Allows multiple private devices to access Internet using one public IP.



- Hides private IPs from Internet.
- Improves security and saves IP addresses.

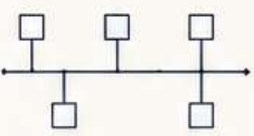
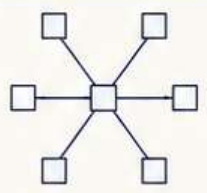
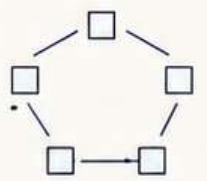
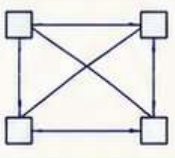
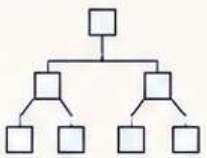
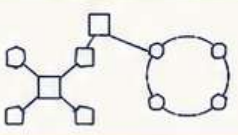
In Short : ARP finds the MAC address, DHCP gives the IP address, and NAT allows private devices to access the Internet.

Page 4 - Network Topologies

→ Network Topology is the physical or logical arrangement of devices (nodes) in a network and how they are connected to each other.

Why Topology is Important ?

- Helps in designing a network.
- Affects performance and reliability.
- Helps in troubleshooting.
- Decides scalability and cost.

Topology	Diagram	Description	Advantages	Disadvantages
Bus		All devices are connected to a single central cable.	• Easy to install • Low cost • Less cable required	• Cable failure stops entire network • Difficult to troubleshoot • Not scalable
Star		All devices are connected to a central device (Hub/Switch).	• Easy to manage • Fault in one device does not affect others • Easy to add/remove devices	• Central device failure stops entire network • More cable required • Costlier than Bus
Ring		Each device is connected to two other devices, forming a circle.	• Data flows in one direction (orderly) • No collisions	• One break can stop entire network • Hard to troubleshoot • Adding/removing devices is difficult
Mesh		Each device is connected to every other device.	• Highly reliable • Fault tolerant • High security	• Expensive • Complex installation • Requires a lot of cables
Tree		Combination of Star topology in a hierarchical structure.	• Easy to expand • Easy to manage • Fault isolation	• Higher cost • Complex cabling • Central root failure affects network
Hybrid		Combination of two or more topologies.	• Flexible • Scalable • More reliable	• Expensive • Complex design • Difficult maintenance

Key Points

- Star topology is most widely used in modern networks (LAN).
- Bus topology is rarely used now.
- Mesh topology is used where reliability and security are critical.
- Hybrid topology is used in large and complex organizations.

Where are they used ?

- Bus → Early Ethernet networks
- Star → Offices, Schools, LANs
- Ring → Metro networks, Token Ring
- Mesh → Military, WAN backbone
- Tree → Organization network
- Hybrid → Internet, Large organizations

In Short : Network Topology decides how devices are connected in a network.
Choosing the right topology improves performance, reliability and scalability.

Page 5 - Proxy, Reverse Proxy & CDN

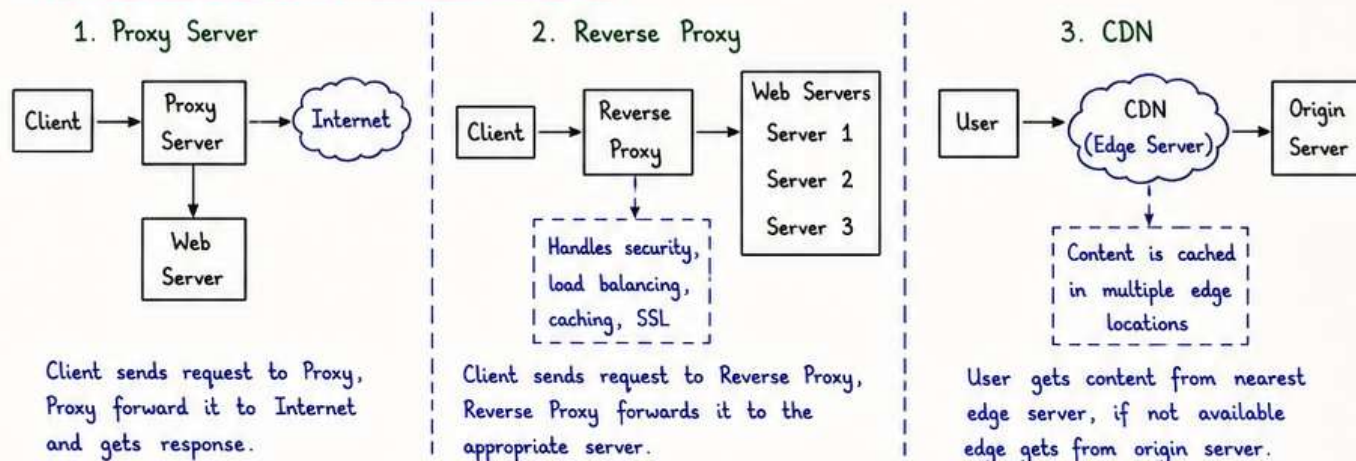
→ These technologies improve performance, security and scalability of applications by acting as intermediaries in the communication between client and server.

What we will cover ?

- Proxy Server
- Reverse Proxy
- CDN (Content Delivery Network)
- Comparison

Feature	Proxy Server	Reverse Proxy	CDN (Content Delivery Network)
Works At	Client Side	Server Side	Distributed Network
Sits Between	Client and Internet	Client and Web Server	User and Origin Server
Main Purpose	Hides client, controls access, caching	Hides server, load balancing, security, SSL termination	Delivers content faster from nearby servers (edge)
Example Use	Company proxy, Web filtering, Caching	Nginx, HAProxy, Traefik	Cloudflare, Akamai, AWS CloudFront
Handles	Outgoing requests from client	Incoming requests to server	Static content (images, videos, CSS, JS, etc.)
Improves	Security, Privacy, Bandwidth saving	Security, Load balancing, SSL offloading	Speed, Performance, Availability

How they work ? (High Level Architecture)



Proxy vs Reverse Proxy vs CDN

Technology	Hides Client	Hides Server	Improves Speed	Load Balancing
Proxy Server	✓	✗	✓	✗
Reverse Proxy	✗	✓	✓	✓
CDN	✗	✗	✓	✓

Key Points

- Proxy is used by clients.
- Reverse Proxy is used by servers.
- CDN stores static content at edge locations.
- All three improve performance, security and scalability in different ways.

In Short : Proxy, Reverse Proxy and CDN reduce load, improve speed, increase security and deliver a better user experience.

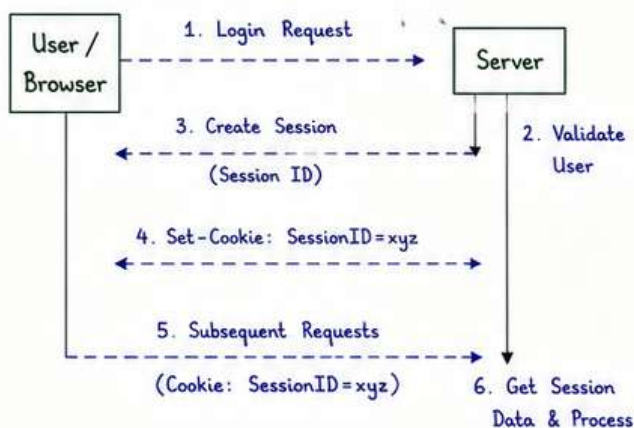
→ These are mechanisms used to maintain state and authenticate users in web applications.

What we will cover ?

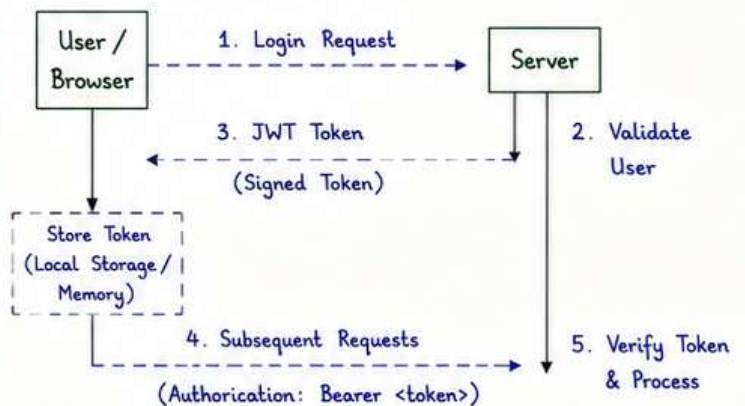
- Cookie
- Session
- JWT (JSON Web Token)
- Comparison
- Login Flow

Feature	Cookie	Session	JWT (JSON Web Token)
What is it ?	Small data stored on client (browser).	Data stored on server and identified by session id.	Token generated by server and stored on client.
Where is data stored ?	Client (Browser)	Server	Client (Browser / Local Storage)
How it works ?	Browser sends cookie with every request to server.	Browser sends session id (in cookie) with request.	Browser sends token in Header with every request.
Size	~ 4 KB	Depends on server configuration	Can be larger (contains claims)
Security	Less secure (can be stolen / modified)	More secure (data on server)	Secure (signed & verified, but visible on client)
Server Load	Low	High (server stores session data)	Low (server is stateless)
Scalability	High	Difficult (needs session sync / sticky session)	High (stateless)
Use Cases	Preferences, theme, tracking, remember me	Login sessions, shopping cart, user session	API Authentication, Microservices, Mobile Apps
Expiry	Expires (set by server)	Expires (server timeout)	Expires (embedded in token)

Login Flow (Session based)



Login Flow (JWT based)



Key Points

- Cookies are key-value pairs stored in browser.
- Sessions store data on server and use session id in cookie.
- JWT is a token (header.payload.signature) used for stateless authentication.
- JWT is widely used in APIs and Microservices.
- Always use HTTPS and HttpOnly cookies for better security.

When to use what ?

- Cookie → Small data, preferences
- Session → Traditional web apps
- JWT → APIs, Mobile apps, Microservices

In Short : Cookies store small data on client, Sessions store data on server and JWT is a token used for secure and stateless authentication.

Page 7 - VPN & Firewall

- VPN and Firewall help in securing network communication and protecting systems from unauthorized access.

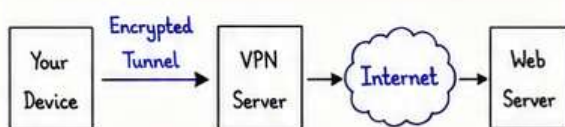
What we will cover ?

- What is VPN ?
- VPN Tunnel
- What is Firewall ?
- Types of Firewall

1. VPN (Virtual Private Network)

- VPN allows a user to connect to a private network over the Internet securely.
- It creates an encrypted tunnel between user and VPN server.

VPN Tunnel



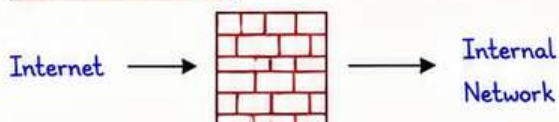
How it Works ?

Your device connects to VPN Server. All traffic goes through encrypted tunnel. Internet sees VPN Server IP, not your real IP.

2. Firewall

- Firewall is a security system that monitors and controls incoming and outgoing network traffic based on rules.
- It acts as a barrier between trusted internal network and untrusted networks (Internet).

Firewall Working



How it Works ?

Firewall checks each packet against rules. If allowed → it forwards. If not allowed → it blocks.

Types of Firewall

Type	Description	Works At	Example / Use
Packet Filtering Firewall	Allows or blocks packets based on IP, Port, Protocol.	Network Layer (L3)	Routers, ACL
Stateful Inspection Firewall	Tracks connection state and allows traffic based on context.	Transport Layer (L4)	Enterprise Firewalls
Application Layer Firewall	Inspects application data and allows/block based on application rules.	Application Layer (L7)	Web Application Firewall (WAF)
Next Generation Firewall (NGFW)	Combines stateful inspection, application awareness, IDS/IPS, and more.	Multi-Layer	Palo Alto, Fortinet

Key Points

- VPN hides your real IP and encrypts data.
- Firewall protects network from unauthorized access.
- Use strong rules in Firewall to allow only necessary traffic.
- VPN is for remote access, Firewall is for network protection.

VPN vs Firewall

Feature	VPN	Firewall
Purpose	Secure remote access	Protect network
Works On	User side	Network boundary
Main Function	Encrypts data	Filters & blocks traffic
Example	OpenVPN, WireGuard	Cisco ASA, FortiGate

In Short : VPN gives secure access to users, Firewall protects your network by allowing only trusted traffic.

Page 8 - WebSocket, Polling & SSE

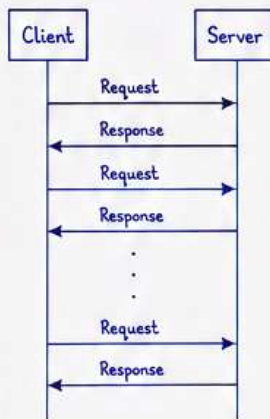
→ These techniques are used to send **real-time** data from server to client.

What we will cover ?

- Polling
- Long Polling
- Server Sent Events (SSE)
- WebSocket
- Comparison

1. Polling

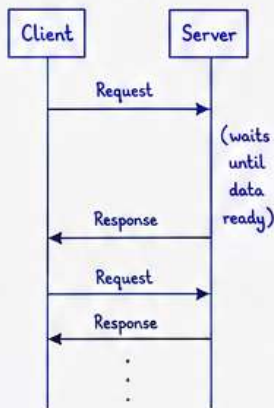
- Client sends request repeatedly to server after fixed interval.
- Simple but not efficient.



Use Case
News Feed, Notifications
(low real-time need)

2. Long Polling

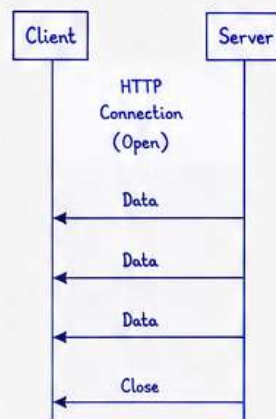
- Client sends request and server holds it until data is ready.
- Reduces empty responses.



Use Case
Chat apps, Live updates

3. Server Sent Events (SSE)

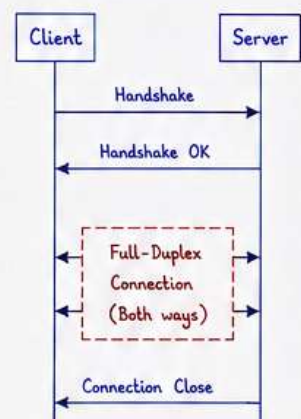
- Server pushes data to client over HTTP.
- One-way communication (Server → Client).



Use Case
Live scores, Stock updates, Alerts

4. WebSocket

- Full-duplex connection over a single TCP connection.
- Real-time two-way communication.



Use Case
Chat apps, Online gaming, Live collaboration

Comparison

Feature	Polling	Long Polling	SSE	WebSocket
Real-time	Low	Medium	High	Very High
Direction	Client ↔ Server (Request based)	Client ↔ Server (Server waits)	Server → Client (One way)	Both ways (Full Duplex)
Connection	Multiple	Multiple (but fewer)	Single (HTTP)	Single (TCP)
Overhead	High	Medium	Low	Very Low
Complexity	Very Simple	Moderate	Simple	Complex
Best For	Low real-time apps	Real-time with less load	One-way live updates	Two-way real-time communication

Examples

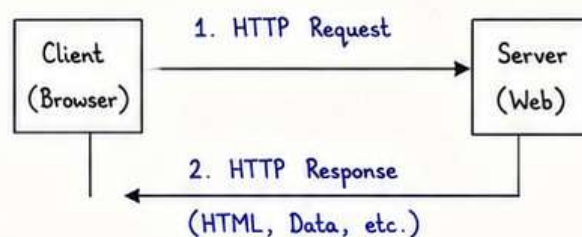
- WebSocket → WhatsApp Web, Online games, Live chat
- SSE → Live cricket score, Stock price updates, System alerts
- Polling → Traditional dashboards, Old applications
- Long Polling → Gmail chat (older version), Live notifications

Page 9 - HTTP, HTTPS & Summary

1. HTTP vs HTTPS

Feature	HTTP	HTTPS
Full Form	HyperText Transfer Protocol	HyperText Transfer Protocol Secure
Security	Not Secure	Secure (Uses SSL/TLS)
Port	80	443
URL	http://	https://
Encryption	No	Yes
Use	Old websites, Internal apps	Websites, Banking, APIs, E-commerce

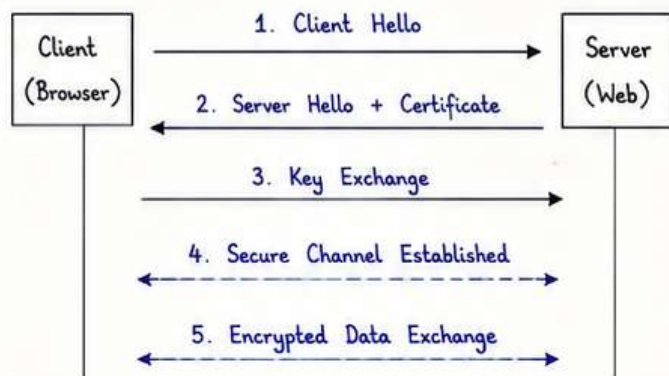
2. HTTP Flow



Note :

- Connection is not encrypted.
- Data can be read by anyone in between (MITM Attack possible).
- Used when security is not required.

3. HTTPS Flow



Note :

- Data is encrypted using SSL/TLS.
- Protects data from eavesdropping and tampering.
- Shows lock  in browser.

4. Summary

- Networks connect devices and help them communicate.
- Topologies decide how devices are arranged.
- Protocols (TCP/IP) make communication possible.
- ARP, DHCP, NAT help in addressing and communication inside network.
- Proxy, Cookies, Sessions, JWT help in web applications.
- VPN and Firewall secure the network.
- Polling, SSE and WebSocket are used for real-time communication.
- Always prefer HTTPS over HTTP.

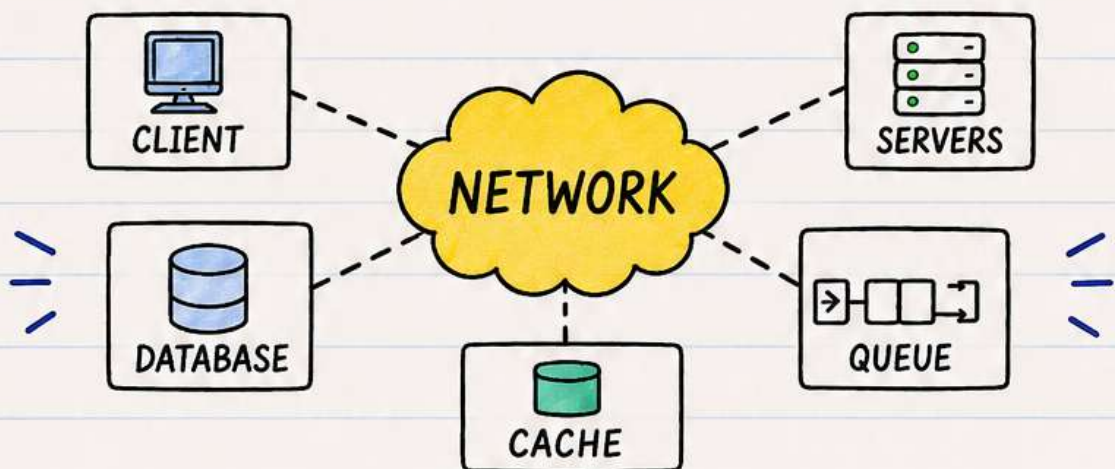
In Short : Computer Networks is all about how data is transferred, how devices communicate and how security is maintained in the modern world.

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